RAGree Choice / SPARROW

An Evidence-Gated Preferential AutoJudge

Abstract. RAGree Choice is a preferential AutoJudge for selecting a winner, top-k, or acceptable response from a candidate pool. Its SPARROW pipeline applies a schema-valid pointwise evidence gate before acquiring hard pairwise ballots, fits a weighted Davidson–Bradley–Terry model with explicit ties, directs additional comparisons toward an uncertain top-k boundary, selectively reverses answer position, and audits the operational ranking using Condorcet, cycle, and Kemeny–Young diagnostics. The governing rule is: **every judge gets a vote; evidence gets a veto.** A live synthetic smoke run required 15 LLM calls rather than 20 for an ungated comparison baseline and produced stable, interpretable rankings. These measurements establish end-to-end mechanism execution, not real-world judging accuracy.

1. Theoretical framework

Preference models with ties. Bradley–Terry models represent pairwise choice through latent utilities; Davidson's extension assigns positive probability to ties rather than treating them as half-observed wins. This is appropriate for RAG evaluation, where materially indistinguishable answers are common and a forced directional verdict can create spurious separation. SPARROW preserves four hard ballot outcomes—`A`, `B`, `TIE`, and `ABSTAIN`—and excludes abstentions from directional fitting.

Pointwise-first, pairwise-final judging. PairS and PRePair motivate separating absolute inspection from comparative judgment. RAGree Choice first constructs one evidence-grounded brief per candidate and only then presents eligible briefs and answers pairwise. This reduces pairwise inconsistency caused by judges discovering criteria opportunistically, while retaining the greater discriminative power of direct comparison. Eligibility requires no critical error and factual-support and citation-validity estimates at or above their configured thresholds. Completeness contributes to `POINTWISE_EVIDENCE_SCORE` but is not itself a gate threshold.

Social choice and auditing. A strict Condorcet winner defeats every other eligible candidate in the observed weighted preference matrix. Condorcet cycles expose collective inconsistency, while a Kemeny–Young order maximizes agreement with all directed margins. Exact Kemeny optimization is NP-hard and requires factorial enumeration in this implementation; it is therefore an audit, not the live rule. Exact search is limited to at most eight candidates, after which a Davidson–BT-seeded adjacent-swap local search is diagnostic only. A Davidson–BT/Kemeny winner disagreement sets `RANKING_STABLE` to zero and signals targeted acquisition or review.

Judge reliability and effective panels. Reliability-aware BT variants such as BT-σ model judge-specific noise or confidence rather than assuming exchangeable voters. Related panel-of-LLMs studies find that a nominal panel often contains far fewer effective-independent votes than its raw judge count: shared training data, model lineage, prompts, and systematic biases induce correlated errors. Consequently, the ballot schema retains `judge_id` and a positive `weight`, and aggregation is judge-weighted and multi-family-ready. The current AutoJudge prototype nevertheless uses only its one injected LLM endpoint; repeated calls are not misrepresented as an independent jury.

2. Design and implementation



Figure 1: SPARROW pipeline. Evidence eligibility precedes preference acquisition; social-choice rules independently audit the fitted live ranking.

2.1. Davidson–BT aggregation

For utilities u_a, u_b, let $\delta = u_a - u_b$ and tie factor $\tau > 0$. The implementation uses

$$D = e^{\delta} + 1 + \tau e^{\frac{\delta}{2}},$$

$$P(A \text{ beats } B) = \frac{e^{\delta}}{D}, \quad P(B \text{ beats } A) = \frac{1}{D}, \quad P(\text{tie}) = \tau\frac{e^{\frac{\delta}{2}}}{D}.$$

The fit is weighted MAP gradient ascent with an L2 prior toward zero. Utilities are identified only up to an additive constant, so every iteration subtracts their mean. For numerical stability, the implemented utility difference is clipped to the interval $[-30, 30]$. `CHOICE_SCORE` is 100 times the mean fitted probability of beating a uniformly selected eligible opponent; it is pool-relative and is not comparable across changing pools without frozen anchors. `PAIRWISE_WIN_RATE` is the observed weighted fraction of non-abstaining ballots won, with ties contributing one half.

2.2. Acquisition, failure semantics, and reversal

Pools of at most five eligible candidates use a full round robin. Larger pools begin with a deterministic cycle, guaranteeing connectivity, then greedily add the not-yet-connected pair having minimum summed current degree until $\lceil n\frac{d}{2} \rceil$ edges are present. Requested degrees below two mean the minimum connected degree of two. An adaptive max-heap then ranks unseen pairs by

$$\frac{2I[\text{crosses top-}\ k\ \text{ boundary}] + \frac{1}{1+|u_a - u_b|}}{\text{expected cost}},$$

concentrating calls on uncertain comparisons capable of moving the decision.

A parsed `ABSTAIN` is a completed ballot and consumes an acquisition edge; transport or schema failure creates no ballot and remains schedulable after bounded retry. Likewise, a missing or malformed evidence brief yields `EVIDENCE_EVALUATED = 0`, not an inferred factual score or gate failure. `NO_ACCEPTABLE_ANSWER = 1` is emitted only when every candidate has a schema-valid brief and every candidate fails eligibility.

Selective position reversal rechecks, within budget, directional ballots crossing the top-k boundary first and then `TIE` ballots; existing `ABSTAIN`s are never reversed. The answer order is swapped and the returned vote is mapped back to the original orientation. Agreement preserves the single original ballot. Genuine disagreement replaces it with one `ABSTAIN`, not a `TIE`, because position sensitivity is uncertainty about direction rather than evidence of equality. A failed reversal leaves the ballot unchanged and does not increment `POSITION_REVERSAL_CHECKED`.

3. Empirical results and analysis

n	Degree	Edges	Full RR	% full	Max spread
10	2	10	45	22.2%	0
10	3	15	45	33.3%	2
10	4	20	45	44.4%	0
10	5	25	45	55.6%	2
16	2	16	120	13.3%	0
16	3	24	120	20.0%	0
16	4 (default)	32	120	26.7%	2
16	5	40	120	33.3%	2

Table 1: Measured sparse-seed acquisition. The $n = 16$, degree-4 result reproduces the design proposal's 32-edge arithmetic.

Table 1 confirms that the connected seed graph is substantially smaller than a full round robin while remaining near-regular. Across the separate sweep $n = 6..29$ and degree $2..7$, the worst observed degree spread was two. Exact regularity was intentionally rejected as cosmetic complexity: connectivity is the correctness requirement, and it holds for every nonnegative requested degree.

Topic / run	Eligible	Pointwise	Win rate	Choice	Condorcet	Stable
cloud / run1	1.0	0.933	1.0	88.01	1.0	1.0
cloud / run2	1.0	0.833	0.5	41.79	0.0	1.0
cloud / run3	0.0	0.600	0.0	0.00	0.0	0.0
cloud / run4	1.0	0.457	0.0	1.77	0.0	1.0
leaf / run1	1.0	0.967	1.0	85.29	1.0	1.0
leaf / run2	1.0	0.783	0.0	1.92	0.0	1.0
leaf / run3	0.0	0.450	0.0	0.00	0.0	0.0
leaf / run4	0.0	0.233	0.0	0.00	0.0	0.0

Table 2: Live RAGree Choice output on the synthetic `kiddie` smoke-test data. Choice is `CHOICE_SCORE`; Pointwise is `POINTWISE_EVIDENCE_SCORE`.

The live run contained two topics, four competing runs per topic, and eight responses. It made eight unconditional evidence-brief calls, four initial pairwise calls among eligible candidates, and three reversal calls, for 15 calls total. An ungated baseline would compare all six pairs per topic: 12 pairwise calls plus the same eight briefs, or 20 total. Thus gating reduced pairwise volume by 66.7% and overall calls by 25% in this small case. No adaptive calls were needed because each eligible pool was already exhausted by its small-pool round robin.

Table 2 illustrates the veto semantics. `run3` has nontrivial pointwise scores on both topics but is ineligible and therefore receives no preference wins. `run1` is the strict Condorcet winner and has the highest pointwise score on both topics; Davidson–BT and Kemeny agree on each eligible winner. `CYCLE_RATE` was 0.0 throughout. Position reversal checked two cloud ballots and one leaf ballot, with disagreement rate 0.0, meaning no position inconsistency was detected among the checked comparisons—not that reversal was omitted.

The `kiddie` collection is a synthetic pipeline smoke test, not a real quality benchmark. These values demonstrate that gate, ballot acquisition, fitting, reconciliation, audit, and leaderboard emission execute end to end and produce differentiated outputs. They do **not** demonstrate real-world judge accuracy or human alignment. At report time, 39 regression tests passed across pure ranking primitives, configuration and failure handling, and an offline end-to-end integration test including eligible, ineligible, unavailable, and reversal-reconciliation cases.

4. Limitations

The prototype has one injected LLM endpoint and therefore no genuine multi-model jury, despite its judge-weighted, multi-family-ready ballot model. Its pointwise scores are LLM estimates, not gold claim-level precision or recall; no authoritative-corpus or retrieval calibration is supplied. There is also no human calibration of eligibility thresholds, utility scale, judge reliability, correlated error, or position-bias rates. Pool-relative scores require frozen reference anchors for longitudinal comparison, and large-pool Kemeny output is only a local-search diagnostic.

5. References

1. Bradley, R. A., and Terry, M. E. **Rank Analysis of Incomplete Block Designs: The Method of Paired Comparisons.**
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4. Condorcet, M. de. Majority preference and the Condorcet winner criterion.
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6. Reliability-aware Bradley–Terry extensions, including **BT-σ**, for judge-specific uncertainty and heterogeneous evaluator quality.
7. **Panel of LLM Evaluators (PoLL)** and related diverse-model jury studies: correlated model judgments and the gap between nominal and effective-independent panel size.
8. RAGree Choice design summary and `sparrow.py` implementation: evidence gating, sparse acquisition, weighted Davidson–BT fitting, selective position reversal, and Condorcet/Kemeny auditing.